

Round 1 - Machine Learning Engineer - {Aditya Maurya}

Round 1 - Online assessment for Machine Learning Engineer role at [FOG](#).

Candidate Name: Aditya Maurya

Form Link: <https://forms.gle/D5DAQefZKHHBCEyv6>

Instructions -

- The assessment should be done solely by yourself.
- Submit the response to assessment within 3 days.

Tips to succeed -

- Pay attention to detail
- Think user first
- Be creative
- Use AI (Chat GPT, Midjourney, Claude, v0, etc...)

All the best and don't forget to have fun!

Question 1

Convert Raw Human Image into Studio - Quality Portrait.

- You are given raw human portrait images captured in uncontrolled conditions (mobile camera / low light / motion / cluttered background).
- Your task is to convert these raw images into studio-quality portrait images using computer vision and image processing techniques.
- The goal is to simulate a professional studio photo look while maintaining a natural appearance and fast inference performance.

Input

- Raw human portrait images (.jpg / .png)
- Images may contain:
 - Motion blur or slight defocus
 - Noisy or cluttered background
 - Uneven lighting
 - Low contrast

- Minor face artifacts

Expected Output

A studio-quality enhanced portrait image with the following improvements:

Mandatory Enhancements

- Remove motion blur (if present)
- Add background blur (portrait / bokeh effect)
- Improve face clarity
- Enhance sharpness and contrast
- Preserve natural skin texture
- Maintain original facial identity

Answer

1. **GitHub Repository URL:** <https://github.com/Surya1CAadi/FOG>
2. **Demo Video URL:**
https://drive.google.com/file/d/1cfCAF5_CVq6Q59LNDsjQF3Uterc5vsvo/view?usp=sharing

Question 2

Have you built any game or worked on a game/gamification related project before? Tell us a little bit about what you did. Attach screenshots where required.

Answer

No, I haven't worked on any game or gamification-related project yet. Most of my projects so far are focused on web development and automation. But I'm interested in game development and confident I can learn and adapt quickly if given the opportunity.